

Solution: By the Invertible Matrix Theorem and the equivalence of the statements (e) and (h), we know that the given statement is True.

(d) If the equation $Ax=b$ has at least one solution for each b in $\mathbb{R}^n$, then the solution is unique for each b .

Solution: From statement (g) of the Invertible Matrix Theorem and Theorem 5 from Chapter 2, the given statement is True.

(e) If there is a b in $\mathbb{R}^n$ such that the equation $Ax=b$ is inconsistent, then the transformation $x \mapsto Ax$ is not one-to-one.

Solution: From the Invertible Matrix Theorem and the equivalence of the statements (g) and (f), we know that there exists a b in $\mathbb{R}^n$ such that the equation $Ax=b$ has no solution if and only if the transformation $x \mapsto Ax$ is not one-to-one.

(3) An $m \times n$ upper triangular matrix is one whose entries below the main diagonal are 0's (as in Exercise 8). When is a square upper triangular matrix invertible? Justify your answer.

Solution: By the Invertible Matrix Theorem, we know that a square $n \times n$ matrix A is invertible if and only if A has n pivot positions. By Theorem 1 from Chapter 1 and the Row Reduction Algorithm, A can be converted to a unique rref. By defn. of pivot positions, it corresponds to the leading 1 of a row in rref. By condition 4 from rref, we know that each leading entry must be 1. Since each row can have at most one leading entry, a matrix of size $n \times n$ can have at most n pivots. By condition 2 from rref, we know that for each row the leading entry must be in a column to the right of the columns of the leading entries of all rows above it. It's easy to see that a matrix A of size $n \times n$ has n pivot positions if and only if the pivots lie on the main diagonal, i.e. each main diagonal entry of A must be non-zero.

upper triangular
 $\therefore$ A square $n \times n$ matrix A is invertible if and only if each main diagonal entry must be non-zero. II

(4) An $m \times n$ lower triangular matrix is one whose entries above the main diagonal are 0's (as in Exercise 3). When is a square lower triangular